# Daniel GODINES

## PERSONAL DATA


ADDRESS: New Mexico State University, 1780 E University Ave, Las Cruces, NM 88003, USA
PHONE: +1 805 3153183
EMAIL: godines@nmsu.edu


## EDUCATION

JUN 2021 - PRESENT — Doctor of Philosophy in ASTRONOMY, **New Mexico State University**, NM
Advisors: Moire PRESCOTT & Wladimir Lyra

AUG 2014 - MAY 2018 — Bachelor of Arts in PHYSICS, **Bard College**, NY
Thesis: "A Microlensing Detection Algorithm for Wide-Field Surveys"
Advisor: Antonios KONTOS

## PUBLICATIONS

*Total Citations: 54 | h-index: 4 (Source: NASA/ADS)*

### Peer-Reviewed Journal Articles

MAY 2026 — *A Hybrid Ensemble and Deep Learning Framework for Detecting High-Redshift Ly$\alpha$ Blobs in Broadband Surveys*
**Godines, D.**, Prescott, M. K. M., 2026, *PASP*, 138, 054505.

JAN 2026 — *On the Mass Budget Problem of Protoplanetary Disks: Streaming Instability and Optically Thick Emission*
**Godines, D.**, Lyra, W., Ricci, L., Yang, C.-C., Simon, J.B., Lim, J., Carrera, D., 2026, *ApJ*, 997, 192.

OCT 2025 — *Dark classification matters: searching for primordial black holes with LSST*†
Crispim Romão, M., Croon, D., Crossey, B., **Godines, D.**, 2025, *JCAP*, 10, 066.

SEP 2025 — *Anomaly detection to identify transients in LSST time series data*†
Crispim Romão, M., Croon, D., **Godines, D.**, 2025, *MNRAS*, 543, 351.

JUN 2019 — *A Machine Learning Classifier for Microlensing in Wide-Field Surveys*
**Godines, D.**, Bachelet, E., Narayan, G., Street, R. A., 2019, *A&C*, 28, 100298.

†Author list is alphabetical.

### Research Notes

SEP 2025 — *Classifying Microlensing Events from ROME/REA*
Schweitzer, A., Street, R. A., Kruszynska, K., **Godines, D.**, 2025, *RNAAS*, 9, 232.

### Selected Manuscripts in Preparation

2026 — *Accelerating Planet Formation Population Synthesis with Machine Learning Surrogates*
**Godines, D.**, et al., In prep.

2026 — *Early Detection of Stellar and Exotic Microlensing in the Rubin Era*
**Godines, D.**, et al., In prep.

2026 — *Automated Discovery of Ly$\alpha$ Blobs in COSMOS*
**Godines, D.**, et al., In prep.

## Awarded Computing Resources

2026B **Rubin Observatory US Data Facility (USDF)** | PI (5,580 CPU hrs, 150 GPU hrs, 9 TB)
*Scaling Microlensing Discovery: A Community Simulation and Classification Suite for LSST*
Awarded via NOIRLab Resource Allocation Committee (RAC).

## Awarded Telescope Time

2025A **Gemini North Observatory** | PI (27.7 hours, Band 3)
*Automated Identification of Lyman-alpha Blobs in the Early Universe*
Awarded via General Queue (GN-2025A-Q-320); Instrument: GMOS-N.

2024A **Gemini North Observatory** | PI (5.94 hours, Band 2)
*Automated Identification of Lyman-alpha Blobs in the Early Universe*
Awarded via Fast Turnaround (GN-2024A-FT-210); Instrument: GMOS-N.

## Professional Service

2026 **Peer Reviewer** | *The Astronomical Journal*
2025 **Peer Reviewer** | *Nature Astronomy*
2025 **Peer Reviewer** | *Journal of Open Source Software* (JOSS)

## Research Experience

2016 – 2019 **Research Intern** | Las Cumbres Observatory
*Gravitational Microlensing Research with Dr. Rachel A. Street*
Engineered **MicroLIA**, a machine-learning engine for real-time microlensing detection. Successfully integrated the framework into the **Rubin Observatory** alert stream and the **Fink** and **Pitt–Google** alert brokers.

## Selected Open-Source Projects

- **MicroLIA** — Machine-learning algorithm for real-time transient detection. Adopted by the LSST Microlensing Science Collaboration and designed for live alert-broker deployment (e.g., FINK, ANTARES, Pitt–Google). Modular architecture supporting general time-series classification and synthetic light curve generation. Docs | Zenodo
- **pyBIA** — End-to-end pipeline for image-based classification and anomaly detection. Includes segmentation-driven cataloging, morphological feature engineering, unsupervised artifact rejection, and routines for training optimized supervised ML pipelines, including deep-learning models with automated data augmentation. Docs | Zenodo
- **protoRT** — Physics-based 3D radiative transfer framework for synthetic radio emission in planet formation simulations. Computes optical depth and emergent intensity through dust density cubes, supporting self-gravitating mono and polydisperse dust with scattering opacity and DSHARP integration. Includes a mass-excess diagnostic for quantifying observational biases from optically thick regions. Docs | Zenodo
- **rubin-lc-simulator** — Open-source framework for simulating realistic multi-band LSST light curves, accounting for official cadence strategies and photometric noise. Docs

## Presentations

**Contributed Talk**, *3rd PFITS+ Meeting on Hydrodynamics of Circumstellar Disks and Planet Formation*, SETI Institute, Mountain View, CA, USA 2026
"On the Mass Budget Problem: Streaming Instability and Optically Thick Emission"

**Poster**, *50 years of Binaries and Disks: Lubow@75*, UNLV, Las Vegas, NV, USA 2024
"On the Mass Budget Problem of Planet Formation Theory: Streaming Instability and Opti-

cally Thick Regions"

**Contributed Talk**, *ODIN Collaboration Meeting*, KIAS, Seoul, South Korea 2024
"Automated Ly$\alpha$ Blob Detection Using Broadband Imaging"

**Contributed Talk**, *Rare Gems in Big Data*, Tucson, AZ, USA 2024
"Classification of Rare Astrophysical Objects: From Galactic to Extragalactic"

## Scholarships and Awards

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| 2025 - 2026 | Zia Award for Excellence in Research |
| 2023 - 2024 | A. Scott Murrell Memorial Endowed Scholarship |
| 2021 - 2023 | William Webber Voyager Graduate Fellowship |
| 2017 - 2018 | LSSTC Enabling Science Program Award |
| 2014 - 2018 | Distinguished Scientist Scholar Award |

## Selected Technical Skills

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| Programming Proficiency: | Python (Expert), C++, SQL (MySQL), Matlab, Mathematica, HTML |
| Data Science & ML: | TensorFlow, PyTorch, scikit-learn, XGBoost, emcee, astropy, rubin-sim |
| Software & Tools: | PyPI, Git, Docker, Sphinx (Documentation), Zenodo, DS9, Photoshop |
| Operating Systems: | GNU/Linux (Ubuntu/CentOS), Mac OS X |
| Language Proficiency: | English, Spanish (Native), French (Conversational) |

## Technical Training & Observational Experience

Summer 2024 **ZTF Summer School** | California Institute of Technology
Advanced training in ZTF data releases and alert-broker integration for transients.

Summer 2018 **ZTF Summer School** | California Institute of Technology
Intensive program on wide-field survey analysis and automated discovery.

Summer 2018 **Research Visit** | Space Telescope Science Institute (STScI)
Optimization of machine learning pipelines for high-dimensional astrophysical data.

Summer 2017 **CUREA Program** | Mt. Wilson Observatory
Two-week residence in observational stellar and solar physics.

Summer 2017 **Technical Workshop** | National Optical Astronomy Observatory (NOAO)
Training on LSST data infrastructure and real-time alert systems.

Summer 2016 **WIYN 3.5m Observatory** | Kitt Peak National Observatory
Observer for a joint imaging campaign supporting the *Kepler* K2 mission.

Summer 2016 **iPTF Summer School** | California Institute of Technology
Foundational training in Time-Domain Astronomy and survey operations.

## Leadership & Outreach

2024–Present **Board Member** | Santa Barbara County Science & Engineering Fair
Managed logistics and STEM advocacy for grades 6–12 across Santa Barbara and San Luis Obispo counties; coordinating pathways to State finals.

June 2024 **Invited Lecture** | Santa Barbara Salon Series
*"From Starlight to Artificial Intelligence: How We Learn to Read the Universe"* (Public Talk).

June 2024 **Invited Lecture** | Santa Barbara Institute of World Culture
*"Cosmic Revelations: Exploring Astronomy's Frontiers"* (Public Talk).

2022 **Guest Science Columnist** | Las Cruces Sun-News
Authored features on *A.I. in Astronomy* and *The Search for Planet 9*.

2014 **Media Feature** | Santa Barbara Independent
Profiled in cover story *"A Comet in Santa Barbara's Astronomical World."*

2013–2014 **Planetary Science Tutor** | Santa Barbara City College
Youngest appointed tutor for undergraduate planetary science course.